Interpret Division of Fractions Flagship

Lesson 2-1-flagship

Name: ______ Date: ______ Class: ______

EVIDENCE LAB MISSION

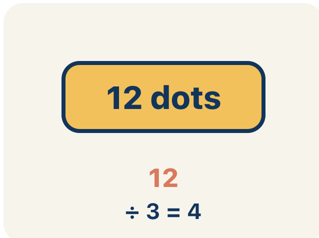
The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact 1/4-foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.

Key Vocabulary

Level 1 support

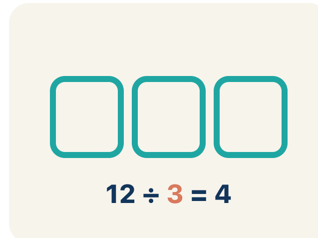
Picture first, then the word, then a plain-language meaning. Say each word out loud.



In 3 ÷ 1/4 = 12, the dividend is 3 — it is the total being split

Dividend

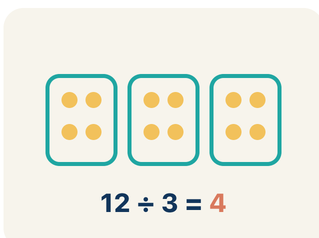
The number you are splitting up in a division problem.



In 3 ÷ 1/4 = 12, the divisor is 1/4 — it is the size of each piece

Divisor

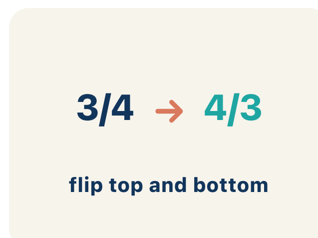
The number you split by in a division problem.



In 3 ÷ 1/4 = 12, the quotient is 12 — there are 12 quarter-size pieces in 3

Quotient

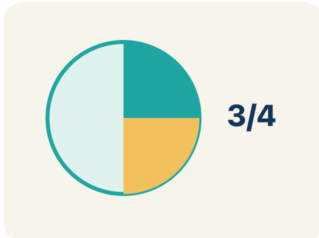
The answer when you divide.



The reciprocal of 1/4 is 4/1 = 4. Multiplying by the reciprocal gives the same result as dividing.

Reciprocal

A fraction turned upside down.



1/2, 1/3, 1/4, 1/5 — each represents one equal part of a whole

Unit fraction

A fraction with 1 on top, like 1/4.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can use models to interpret what it means to divide by a fraction.

1 Notice

Agent Reyes finds a 3-foot strip of tape at the crime scene.

2 Key idea

I can use models to interpret what it means to divide by a fraction.

3 Words I'll use

Dividend **Divisor** **Quotient** **Reciprocal** **Unit fraction**

🤔 Think About It

- What total length is being divided?
- What is the size of each section?
- Will the answer be greater than or less than 3?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: What does 3 ÷ 1/4 mean?

A. How many 1/4-size pieces fit into 3

B. 3 groups of 1/4

C. 3 minus 1/4

D. 1/4 of 3

Step 1 3 ÷ 1/4 asks: how many 1/4-size pieces fit into 3?

Step 2 The answer is 12, because each whole has 4 fourths, and 3 × 4 = 12.

✅ **Answer:** A. How many 1/4-size pieces fit into 3

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which expression means 'how many 1/3-size pieces are in 2'?

A. $2 \div 1/3$

B. $2 \times 1/3$

C. $1/3 \div 2$

D. $2 + 1/3$

Step 1 ______

Step 2 ______

Answer: ______

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: A rope is 4 feet long. How many 1/2-foot pieces can be cut from it?

A. 8 pieces

B. 2 pieces

C. 4 pieces

D. 1/2 piece

Show your work:

______

______

______

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A 3-foot strip of evidence tape must be cut into 1/4-foot sections. Before calculating, will the number of pieces be more or fewer than 3? Explain your thinking.

Level 1 support

Start here: Because each piece is smaller than a whole foot, 3 feet makes more than 3 pieces.

💡 **Need a hint?**

Sentence starters (optional)

There will be ____ than 3 pieces because each piece is ____ than 1 foot.

Habrá ____ de 3 pedazos porque cada pedazo es ____ de 1 pie.

Dividing 3 by 1/4 means asking how many ____ fit into ____.

Dividir 3 entre 1/4 significa preguntar cuántos ____ caben en ____.

Word bank: dividend | divisor | quotient | unit fraction | section

Listen for: *Students reason that since 1/4 is less than 1, more than 3 pieces fit, so the quotient is greater than the dividend.*

Level 2

Why does dividing by a unit fraction like 1/4 give an answer larger than the number you started with? Justify the idea, not just this case.

- Dividing by a number less than 1 makes the answer ____ because ____.
- This always happens when the divisor is ____.

EXPLORE

On your fraction bar, how many 1/4-foot sections fit into one whole foot? Point to where you see the pieces.

Level 1 support

Start here: Four 1/4-foot sections tile one whole foot, so each foot holds 4 pieces.

💡 **Need a hint?**

Sentence starters (optional)

In one whole foot I see ___ sections of 1/4 each.

En un pie completo veo ___ secciones de 1/4 cada una.

So 3 feet hold ___ pieces because 3 times ___ equals ___.

Entonces 3 pies tienen ___ pedazos porque 3 por ___ es ___.

Word bank: dividend | divisor | quotient | unit fraction | model

Listen for: *Students see 4 pieces per foot and connect it to $3 \times 4 = 12$ sections, matching $3 \div 1/4 = 12$.*

Level 2

Use the reciprocal to show 3 divided by 1/4 the algorithm way. Why does multiplying by 4 give the same answer as the bar model?

- Using the reciprocal, 3 times ___ equals ___ because ___.
- Multiplying by the reciprocal works because ___.

CONNECT

Where else outside the lab would you divide a whole number by a unit fraction, and how would you set it up?

Level 1 support

Start here: Any time you cut a whole amount into equal fractional-size pieces, you divide by a fraction.

💡 **Need a hint?**

Sentence starters (optional)

I would divide a whole number by a unit fraction when ____.

Dividiría un número entero entre una fracción unitaria cuando ____.

I would set it up as ____ divided by ____.

Lo plantearía como ____ entre ____.

Word bank: **dividend** **divisor** **quotient** **reciprocal** **unit fraction**

Listen for: *Students give a real example (ribbon cut into 1/3-foot pieces, juice into 1/2-cup servings) and name the dividend and divisor.*

Level 2

Write your own crime-scene word problem that needs dividing a whole number by a unit fraction, and explain what the quotient means in your story.

- My problem is ____, and the quotient tells how many ____.
- I know it needs division by a fraction because ____.

Write About the Math

The Writing Revolution

I can explain a fraction division using the words dividend, divisor, quotient, reciprocal, and unit fraction.

1. Kernel Sentence

subject + verb

Model: Dividend is the number you are splitting up in a division problem.
Dividendo es el número que estás repartiendo en una división.

Write a kernel sentence about dividend. Use a subject and a verb.
Escribe una oración base sobre dividendo. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Dividend matters in math
Dividendo importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Dividend matters in math because ____.** *Dividendo importa en matemáticas porque ____.*
but *pero*	**Dividend matters in math, but ____.** *Dividendo importa en matemáticas, pero ____.*
so *entonces*	**Dividend matters in math, so ____.** *Dividendo importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about dividend.
Di un hecho verdadero sobre dividend.

Dividend ____.

Question
Pregunta

Ask a question about dividend.
Haz una pregunta sobre dividend.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about dividend.
Muestra entusiasmo sobre dividend.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with dividend.
Dile a un compañero qué hacer con dividend.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

There will be ____ than 3 pieces because each piece is ____ than 1 foot.
Habrá ____ de 3 pedazos porque cada pedazo es ____ de 1 pie.

Dividing 3 by 1/4 means asking how many ____ fit into ____.
Dividir 3 entre 1/4 significa preguntar cuántos ____ caben en ____.

In one whole foot I see ____ sections of 1/4 each.
En un pie completo veo ____ secciones de 1/4 cada una.

Try It

Solve on your own. Check the answer key when you are done.

1. What is 6 ÷ 1/5?

A. 30

B. 6/5

C. 5/6

D. 1

Show your work:

2. A shelf is 5 feet long. Books are each 1/4 foot wide. How many books fit on the shelf? Which expression represents this?

A. 5 ÷ 1/4 = 20 books

B. 5 × 1/4 = 1 1/4 books

C. 1/4 ÷ 5 = 1/20 book

D. 5 + 1/4 = 5 1/4 books

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Dividing 6 by 1/2 gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number (1/2) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

Sentence starter: When I divide 6 by 2, I am ____. When I divide 6 by 1/2, I am ____. The answer is bigger because ____.

Show your work:

Reflect — Exit Ticket

What does 5 ÷ 1/4 mean, and what is the quotient?

A. How many 1/4-size pieces fit into 5; quotient is 20

B. 5 groups of 1/4; quotient is 5/4

C. 1/4 of 5; quotient is 5/4

D. 5 minus 1/4; quotient is 4 3/4

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 2 ÷ 1/3 *— Finding how many 1/3-size pieces fit into 2 is represented by 2 ÷ 1/3. The answer is 6.*
2. **You Try (notes):** A. 8 pieces *— 4 ÷ 1/2 = 4 × 2 = 8 pieces. Each whole foot contains 2 halves, and 4 × 2 = 8.*
3. **Try It 1:** A. 30 *— 6 ÷ 1/5 means how many 1/5-size pieces fit into 6. Each whole has 5 fifths, so 6 × 5 = 30.*
4. **Try It 2:** A. 5 ÷ 1/4 = 20 books *— You are finding how many 1/4-foot pieces fit into 5 feet, which is 5 ÷ 1/4 = 5 × 4 = 20 books.*
5. **Exit Ticket:** A. How many 1/4-size pieces fit into 5; quotient is 20 *— 5 ÷ 1/4 means 'how many 1/4-size pieces fit into 5.' Since each whole has 4 fourths, 5 × 4 = 20.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Dividend is the number you are splitting up in a division problem.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).